

2.4 Electric Circuits

This topic covers the definitions of various electrical quantities, for example current, potential difference and resistance, Ohm's law and non-ohmic conductors, potential dividers, e.m.f. and internal resistance of cells and negative temperature coefficient thermistors.

This topic should be studied using applications such as space technology.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiments.

Candidates will be assessed on their ability to:

64	understand that electric current is the rate of flow of charged particles and be able to use the equation $I = \frac{\Delta Q}{\Delta t}$
1A = 1C/s	Electric current is the rate of flow of charged particles (through a conductor) In a metal wire, these are negatively charged electrons, while in other materials like electrolytes or plasmas, they can be ions and electrons Analogy (to better understand current) Think of electric current as the amount of water flowing through a pipe per second In the case of an electric current, the "water" is the charged particles, and the "pipe" is the conductor like a wire
65	understand how to use the equation $V = \frac{W}{Q}$
1V = 1J/C	Voltage is the energy transferred per unit charge This is a.k.a. potential difference as it is the difference in electric potential energy between two points, representing the work or energy needed to move a unit of electric charge between those two points Analogy (to better understand voltage) Think of voltage (potential difference) as the water pressure in a pipe that is pushing water particles to flow and do work (often as a result of a difference in gravitational potential energy) In the case of an electric circuit, the voltage is the electrical "pressure" or the difference in electrical potential energy between two points in a circuit

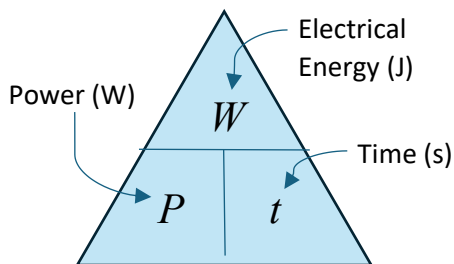
66	understand that resistance is defined by $R = \frac{V}{I}$ and that Ohm's law is a special case when $I \propto V$ for constant temperature
	Resistance is the opposition to the flow of current The equation for resistance $R = V/I$ is true for both ohmic and non-ohmic components so can be used to calculate resistance in any situation Ohm's Law Ohm's law states that <i>"the current through a component is directly proportional to the voltage across it, provided that the temperature remains constant"</i> $V = IR$ A graph of current against voltage for an ohmic material would be a straight line through origin where the gradient of graph represents constant resistance Ohmic vs. non-ohmic materials Ohmic materials obey Ohm's law and have a constant resistance (as long as temperature is controlled) Examples include most metallic conductors such as copper wire Non-ohmic materials do not obey Ohm's law because their resistance changes with voltage, current, or other physical factors. Examples include semiconductors such as diodes and transistors, filament light bulbs, electrolytes, LDRs, thermistors, etc
67	(a) understand how the distribution of current in a circuit is a consequence of charge conservation (b) understand how the distribution of potential differences in a circuit is a consequence of energy conservation
(a)	Conservation of Charge - The distribution of current in a circuit is a consequence of charge conservation, which states that charge can neither be created nor destroyed, only transferred or redistributed within the system - This means the total amount of charge within a circuit cannot increase or decrease while the circuit is functioning due to conservation of charge In a series circuit, the current is the same at every point as charge cannot accumulate at any point since in a series circuit, all components are connected end-to-end in a single loop In a parallel circuit, the current from the source reaches a junction and divides. At the junction, charge conservation dictates that the total current entering must equal the total current leaving <i>Sum of current entering a junction = Sum of current leaving a junction</i> At a junction, current splits based on the inverse proportion of the resistance of each branch, meaning more current will flow through the path of least resistance

This principle is embodied in Kirchhoff's Junction Rule!

(b)	Conservation of Energy The distribution of potential differences in a circuit is a consequence of energy conservation because the total electrical energy supplied by the power source must be equal to the total energy dissipated by all components in the circuit In a closed loop, the sum of all potential differences (voltages) must be zero Due to the conservation of energy, energy is never used up or lost in a circuit In a series circuit, the total potential difference provided by the battery is divided among the components (like resistors) in the circuit This means, sum of potential differences across each component = total emf In mathematical terms, The sum of voltages across each resistor ($V_1, V_2, V_3, \dots$) equals the total voltage of the supply (V_S): $V_S = V_1 + V_2 + V_3 + \dots$ In a parallel circuit, each branch forms a closed loop The potential difference across each individual branch is the same and is equal to the supply voltage (e.m.f.) In mathematical terms, the voltage across each branch ($V_1, V_2, V_3, \dots$) is the same: $V_1 = V_2 = V_3 = \dots = V_S$												
68	be able to derive the equations for combining resistances in series and parallel using the principles of charge and energy conservation, and be able to use these equations												
	<table><tr><td> Formula Derivation for R_{total} in Series In series circuit, $V_{\text{total}} = V_1 + V_2$ Current is the same throughout $\Rightarrow IR_{\text{total}} = IR_1 + IR_2$ $IR_{\text{total}} = I(R_1 + R_2)$ $R_{\text{total}} = R_1 + R_2$ </td><td> Formula Derivation for R_{total} in Parallel In parallel circuit, $I_{\text{total}} = I_1 + I_2$ $\frac{V}{R_{\text{total}}} = \frac{V_1}{R_1} + \frac{V_2}{R_1}$ Voltage is same in parallel so $\div V \Rightarrow \frac{1}{R_{\text{Total}}} = \frac{1}{R_1} + \frac{1}{R_2}$ </td></tr></table>	Formula Derivation for R_{total} in Series In series circuit, $V_{\text{total}} = V_1 + V_2$ Current is the same throughout $\Rightarrow IR_{\text{total}} = IR_1 + IR_2$ $IR_{\text{total}} = I(R_1 + R_2)$ $R_{\text{total}} = R_1 + R_2$	Formula Derivation for R_{total} in Parallel In parallel circuit, $I_{\text{total}} = I_1 + I_2$ $\frac{V}{R_{\text{total}}} = \frac{V_1}{R_1} + \frac{V_2}{R_1}$ Voltage is same in parallel so $\div V \Rightarrow \frac{1}{R_{\text{Total}}} = \frac{1}{R_1} + \frac{1}{R_2}$										
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69	be able to use the equations $P = VI$, $W = VIt$ and be able to derive and use related equations, e.g. $P = I^2R$ and $P = \frac{V^2}{R}$												
Material with temperature For most conductors, resistance increases as temperature increases, while resistance decreases for semiconductors	Electrical Power Power (W) P Current (A) I Voltage (V) V Electric power (P) is the rate at which energy is transferred in an electrical circuit												

Electrical energy can have the symbol "E" or "W" for Work Done but either is fine!

Electrical Energy



Electric energy (W) is the total energy transferred over a period of time

Since $W = Pt$ and $P = IV$, the equation becomes $W = IVt$

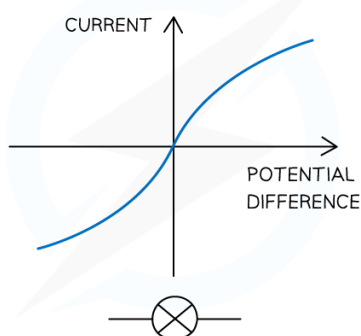
Related Power equations

Using Ohm's law ($V = IR$), you can substitute for either V or I in the base power equation ($P = VI$) to derive two other common power formulas $P = I^2R$ and $P = V^2 / R$. These are especially useful for problems where you are not given all three variables.

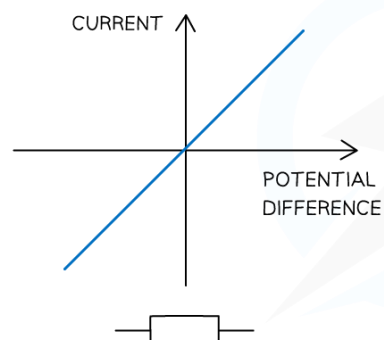
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understand how to sketch, recognise and interpret current-potential difference graphs for components, including ohmic conductors, filament bulbs, thermistors and diodes

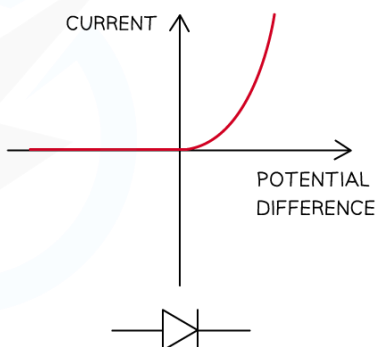
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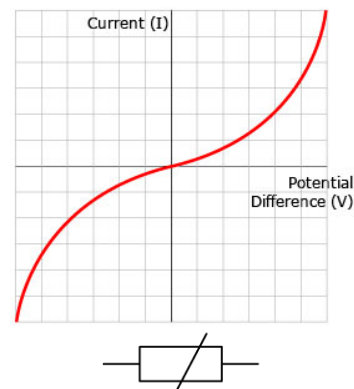
RESISTOR



SEMICONDUCTOR DIODE



THERMISTOR



71

be able to use the equation $R = \frac{\rho L}{A}$

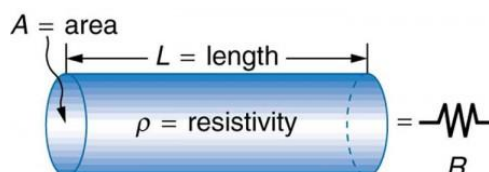
Resistivity (ρ)

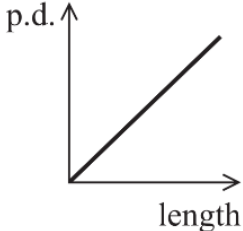
A fundamental property of a **material** that measures its resistance to electric current (how strongly a substance opposes the flow of electrons) (measured in Ωm)

This means resistivity is constant for a particular material regardless of its shape and size unlike resistance which DOES depend on the object's shape and size

$$R = \frac{\rho L}{A}$$

ρ = resistivity
 L = length
 A = cross sectional area

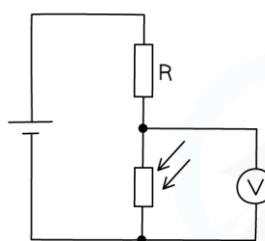
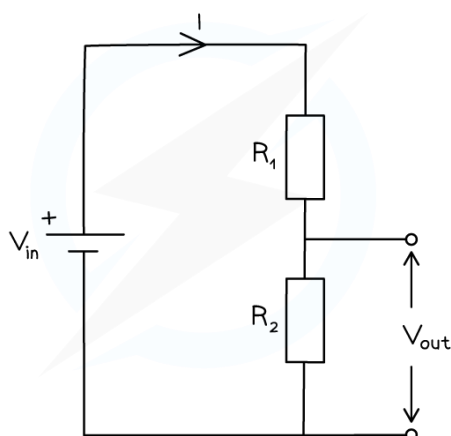


72	CORE PRACTICAL 7: Determine the electrical resistivity of a material
73	be able to use $I = nqvA$ to explain the large range of resistivities of different materials
	The equation $I = nqvA$ can be used to explain the large range of resistivities of different materials Firstly, I = current = flow of charge per unit time, n = number density = number of charge carriers per unit volume, q = charge of a single carrier, v = drift velocity = average speed of the charge carriers as they move through the material, A = cross-sectional area 1. Resistivity in conductors are mostly low because conductors are materials that have a large number of charge carriers (free electrons) meaning it has a large "n" 2. Resistivity in insulators on the other hand are quite high as insulators are materials that have a small number of charge carriers (free electrons) meaning it has a small "n" 3. Resistivity in semiconductors are somewhere in-between as their value of "n" is between conductors and insulators. Their resistivity can be changed by altering conditions like temperature. At higher temperatures, more charge carriers are freed, increasing "n" which decreases resistivity
74	understand how the potential along a uniform current-carrying wire varies with the distance along it
	How the potential varies with length of uniform wire - The potential along a uniform, current-carrying wire decreases linearly (uniformly) with distance from the positive terminal towards the negative terminal - This is because at a constant temperature, the length of a uniform conductor is directly proportional to its resistance - This means as length increases, resistance also increases - Since $V = IR$ and current is constant, p.d. across it also increases  A higher resistance takes a larger share of the voltage, while a lower resistance takes a smaller share

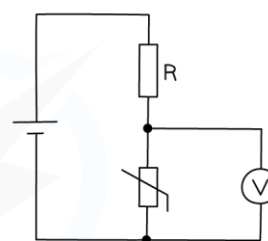
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understand the principles of a potential divider circuit and understand how to calculate potential differences and resistances in such a circuit

POTENTIAL DIVIDER EQUATION: $V_{out} = \frac{R_2}{R_1 + R_2} V_{in}$



LDR POTENTIAL DIVIDER CIRCUIT



THERMISTOR POTENTIAL DIVIDER CIRCUIT

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Potential dividers are circuits that divide voltage of the power supply in a certain ratio using two resistors in series

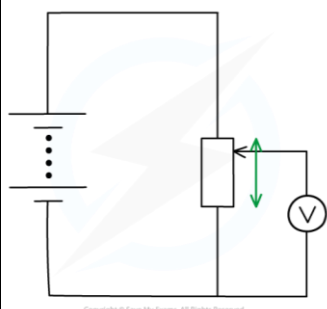
If the resistance of one of the resistors is increased, the potential difference across that resistor will increase whilst the potential difference across the other resistor will decrease

Since current is constant in a series circuit,
We can also link the ratio of p.d. and resistance across each resistor using:

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

76

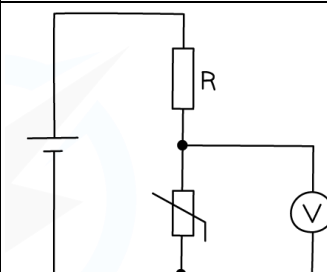
be able to analyse potential divider circuits where one resistance is variable including thermistors and light dependent resistors (LDRs)



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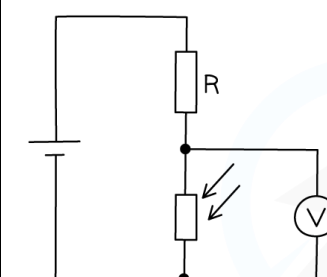
Potentiometer

If the slider in the diagram is moved upwards, the resistance of the lower part will increase and so the potential difference across it will increase



Thermistor

When temperature increases, the resistance of thermistor decreases and so the potential difference across the thermistor also decreases which means the potential difference across the other resistor increases



Light Intensity

When light intensity increases, resistance of LDR decreases and so the potential difference across the LDR also decreases which means the potential difference across the other resistor increases

Just remember!
A higher resistance takes a larger share of the voltage, while a lower resistance takes a smaller share

77 E.m.f. isn't actually a force but rather a measure of energy transferred per charge (measured in volts, V)	know the definition of <i>electromotive force (e.m.f.)</i> and understand what is meant by <i>internal resistance</i> and know how to distinguish between e.m.f. and <i>terminal potential difference</i> Electromotive force is defined as: "The total energy transferred from other forms (chemical, mechanical) into electrical energy by a source per unit charge, representing the total voltage in an open circuit - In essence, electromotive force is the work done moving a unit charge around the whole circuit - Internal resistance is the resistance within a power supply (e.g. battery) causing a drop in voltage when current flows (this is due to resistance offered by the internal materials such as electrodes and electrolyte in a power supply) To find the e.m.f. You need to measure the voltage and current across the power supply when no current flows because e.m.f. is the voltage measured across the terminals of a source when no current flows
78	CORE PRACTICAL 8: Determine the e.m.f. and internal resistance of an electrical cell
79	understand how changes of resistance with temperature may be modelled in terms of lattice vibrations and number of conduction electrons and understand how to apply this model to metallic conductors and negative temperature coefficient thermistors
For metals: The no. of free electrons (n) remains relatively constant, but their mobility decreases with temperature	Changes of resistance with temperature - All materials have some resistance to the flow of charge - This is because as free electrons move through a wire, they collide with the metal ions, losing energy as heat - Changes in resistance with temperature are based on two factors: 1. Lattice vibrations (which happen in both metals and semiconductors) 2. Number of charge carriers (n) (which increase A LOT in semiconductors while stays fairly constant in metals) For metallic conductors (a.k.a. in metals): - As temperature increases, the vibrations of the ions in the lattice also increase which increases the rate of collisions between free electrons and metal ions - Drift velocity of electrons decreases which results in a decrease in current - As $V = IR$, a decrease in current causes an increase in resistance - That is why too high a current will cause an increase in temperature and thus an increase in resistance In general, As temperature increases, resistance (and resistivity) increases and current decreases As temperature decreases, resistance (and resistivity) decreases and current increases

	For semiconductors (negative temperature coefficient thermistors): - As temperature increases, the number density (n) of charge carriers (electrons in the case of wires) increases because in semiconducting materials, increased thermal energy releases a large number of electrons from the valence band into the conduction band - This means the number of conduction electrons per unit volume increases which results in a decrease in resistance (and resistivity) - That is why even though lattice vibrations increases resistance, the substantial increase in the number of charge carriers outweighs the increase in lattice vibration scattering, leading to a net decrease in resistance In general, As temperature increases, resistance (and resistivity) decreases and current decreases As temperature decreases, resistance (and resistivity) increases and current increases
80	understand how changes of resistance with illumination may be modelled in terms of the number of conduction electrons and understand how to apply this model to LDRs
	Changes of resistance with light intensity - LDRs are also made of semiconducting materials - Hence, resistance of LDR decreases with increasing light intensity (or illumination) because absorption of photon by electrons excites them from the valence to conduction band - As the light intensity falling on an LDR increases, more electrons are moved from the valence to the conduction band - This leads to an increase in number of charge carriers per unit volume which in turn results in a decrease in resistivity and the resistance of LDR LDR GRAPH RESISTANCE / Ω LOW LIGHT INTENSITY, HIGH RESISTANCE HIGH LIGHT INTENSITY, LOW RESISTANCE LIGHT INTENSITY / Wm^{-2} LDR CIRCUIT SYMBOL: Copyright © Save My Exams. All Rights Reserved

Electrical circuit symbols

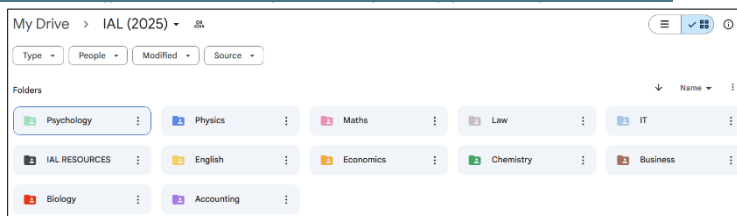
Description	Symbol	Description	Symbol
Conductors crossing with no connection		Heater	
Junction of conductors		Thermistor	
Open switch		Light-dependent resistor (LDR)	
Cell		Diode	
Battery of cells		Light-emitting diode (LED)	
Power supply (DC)		Lamp	
Power supply (AC)		Loudspeaker	
Transformer		Microphone	
Ammeter		Electric bell	
Voltmeter		Earth or ground	
Fixed resistor		Motor	
Variable resistor		Generator	
		Fuse/circuit breaker	

Although these are the forms of circuit symbols that will be used in examination papers, there may be other internationally agreed symbols that are acceptable in student answers.

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